



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

**BE - MCA
I Semester
Dec.2023/Jan.2024
2022 Scheme**

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INDEX

Sl.No.	Sub-Code	Subject Title	Exam Date
1	22MCA11	Mathematical Foundation for Computer Applications	Dec.2023/Jan.2024
2	22MCA12	Operating System Concepts	Dec.2023/Jan.2024
3	22MCA13	Data Structures with Algorithms	Dec.2023/Jan.2024
4	22MCA14	Computer Networks	Dec.2023/Jan.2024
5	22MCA15	Design and Analysis of Algorithm	Dec.2023/Jan.2024
6	22RMI18	Research Methodology and IPR	Dec.2023/Jan.2024

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First Semester MCA Degree Examination, Dec.2023/Jan.2024
Mathematical Foundation for Computer Applications

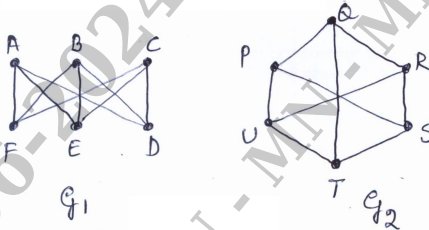
Time: 3 hrs.

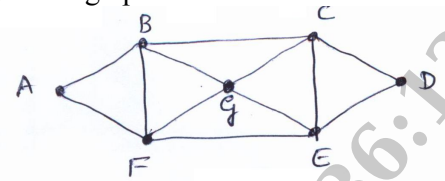
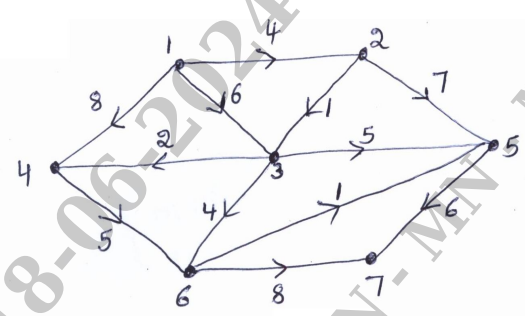
Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	Define a set, an empty set and a singleton set, with an example for each.	6	L3	CO1	
	b.	For any two sets A and B, prove that (i) $\overline{A \cap B} = \overline{A} \cup \overline{B}$. (ii) $\overline{A \cup B} = \overline{A} \cap \overline{B}$.	7	L3	CO1	
	c.	Find the eigen values and eigen vectors of the matrix $\begin{pmatrix} 3 & 2 \\ -1 & 0 \end{pmatrix}$.	7	L3	CO1	
OR						
Q.2	a.	Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 4, 6, 8\}$, $B = \{2, 4, 5, 9\}$. Compute the following : (i) $A \cup B$ (ii) $A \cap B$ (iii) $A - B$ (iv) $\overline{A}$	6	L3	CO1	
	b.	A total of 1232 students have taken a course in Java, 879 in C and 114 in C++. Further, 103 have taken courses in both Java and C, 23 in both Java and C++ and 14 in both C and C++. If 2092 students have taken at least one of Java, C and C++, how many students have taken a course in all three subjects.	7	L3	CO1	
	c.	State pigeon-hole principle. Show that if 50 books in a library contain a total of 27551 pages, one of the books must have atleast 552 pages.	7	L3	CO1	
Module – 2						
Q.3	a.	Write the converse, inverse and contrapositive of, "If 2 is an even number then 7 is a prime number".	6	L2	CO3	
	b.	Define tautology. Show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology.	7	L2	CO3	
	c.	If $A = \{1, 2, 3, 4, 5\}$ is the universe of discourse, determine the truth values of, (i) $\forall x \in A, (x + 2 < 10)$ (ii) $\exists x \in A, (x + 2 = 10)$ (iii) $\forall x \in A, (x^2 \leq 25)$	7	L2	CO3	

OR					
Q.4	a.	Show that for any three propositions p, q, r, $p \rightarrow (q \wedge r) \Leftrightarrow [(p \rightarrow q) \wedge (p \rightarrow r)]$.	6	L2	CO3
	b.	Negate and simplify : (i) $\forall x, [p(x) \wedge \neg q(x)]$. (ii) $\exists x, [p(x) \vee q(x)] \rightarrow r(x)$	7	L2	CO3
	c.	Show that the following argument is valid : “If today is Tuesday, I have a test in Mathematics or Economics. If my Economics professor is sick, I will not have a test in Economics. Today is Tuesday and my Economics professor is sick. Therefore, I have a test in Mathematics”.	7	L2	CO3
Module – 3					
Q.5	a.	Let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by ‘aRb iff a is a multiple of b’. Write down R as a set of ordered pairs. Write matrix and digraph of R.	6	L3	CO1
	b.	Consider $A = \{1, 2, 3, \dots, 11, 12\}$. The relation R on A is defined as $(x, y) \in R$ iff $x - y$ is a multiple of 5. Verify that R is an equivalence relation.	7	L3	CO1
	c.	Let $A = \{2, 3, 4, 6, 8, 12, 24\}$ and R denote the partial order of divisibility ‘xRy iff x divides y’. Find R and draw its Hasse diagram.	7	L3	CO1
OR					
Q.6	a.	Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. Relations R and S from A to B are given by, $M_R = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, M_S = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$ Find (i) $R \cup S$ (ii) $R \cap S$ (iii) $\bar{S}$	6	L3	CO1
	b.	Let $A = \{1, 2, 3, 4\}$ & $R = \{(1, 1), (1, 2), (2, 3), (3, 4)\}$, $S = \{(3, 1), (4, 4), (2, 4), (1, 4)\}$ be relations on A. Determine the relations $R \circ S$, $S \circ R$, R^2 , S^2 and draw their diagrams.	7	L3	CO1
	c.	Consider the POSET whose Hasse diagram Fig. Q6 (c) is as given below. Consider $B = \{3, 4, 5\}$. Find (i) Maximal elements (ii) Minimal elements (iii) Greatest elements (iv) Least elements (v) Upper bound of B (vi) Lower Bounds of B (if they exist)	7	L2	CO4
Fig. Q6 (c)					

Module – 4																							
Q.7	a.	Find the value of K such that the following distribution represents a finite probability distribution. Also find its mean and variance. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>P(x)</td><td>0</td><td>k</td><td>2k</td><td>2k</td><td>3k</td><td>k²</td><td>2k²</td><td>7k²+k</td></tr></table>	x	0	1	2	3	4	5	6	7	P(x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k	10	L3	CO2
x	0	1	2	3	4	5	6	7															
P(x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k															
	b.	When a coin is tossed 4 times, find the probability of getting, (i) exactly one head (ii) atmost 3 heads (iii) atleast two heads.	10	L3	CO2																		
OR																							
Q.8	a.	Find K such that $f(x) = \begin{cases} kx^2, & -3 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$ is a probability density function. Also find (i) $P(1 \leq x \leq 2)$ (ii) $P(x > 1)$	6	L3	CO2																		
	b.	In a certain town, the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that the shower will last for, (i) 10 minutes or more (ii) less than 10 minutes (iii) between 10 and 12 minutes.	7	L3	CO2																		
	c.	The marks of 1000 students in an exam follows a normal distribution with mean 70 and standard deviation 5. Find the number of students whose marks will be, (i) less than 65 (ii) more than 75 (iii) 65 to 75. Given that $\phi(1) = 0.3413$, where $z = \frac{x - \mu}{\sigma}$.	7	L3	CO2																		
Module – 5																							
Q.9	a.	Define the following with an example for each : (i) Simple graph (ii) Regular graph (iii) Spanning subgraph	6	L2	CO4																		
	b.	Check whether the following graphs in Fig. Q9 (b) are isomorphic.  Fig. Q9 (b)	7	L2	CO4																		
	c.	Explain Konigsberg Bridge Problem.	7	L2	CO4																		
OR																							
Q.10	a.	Define Euler circuit, Hamilton cycle and Planar graphs with an example for each.	6	L2	CO4																		

	b. Give graph coloring of the graph F.  Fig. Q10 (b)	7	L2	CO4
	c. Using the Dijkstra's algorithm. Obtain the shortest path from vertex 1 to each of the other vertices in the weighted, directed network shown below in Fig. Q10 (c)  Fig. Q10 (c)	7	L2	CO4

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First Semester MCA Degree Examination, Dec.2023/Jan.2024 Operating System Concepts

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C																																																																					
Q.1	a.	What is an Operating System? Explain with a neat diagram the components of computer systems.	10	L2	CO1																																																																					
	b.	Explain various services provided by the operating system.	10	L2	CO1																																																																					
OR																																																																										
Q.2	a.	Explain with the neat diagram the memory hierarchy.	10	L2	CO1																																																																					
	b.	Explain different types of system program.	10	L2	CO1																																																																					
Module – 2																																																																										
Q.3	a.	Illustrate with a neat sketch, the process state and process control block.	10	L3	CO2																																																																					
	b.	Discuss the various multithreading models. Also mention the benefits of multithreading.	10	L2	CO2																																																																					
OR																																																																										
Q.4	a.	Calculate the Average Waiting Time by drawing Gantt chart for i) Round robin method with the time quantum (q) = 2 ms. <table><tr><td>Process ID</td><td>Arrival Time (AT)</td><td>Burst Time (BT)</td></tr><tr><td>P1</td><td>0</td><td>5</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>2</td></tr><tr><td>P4</td><td>4</td><td>1</td></tr></table> ii) Shortest Remaining Time First (SRTF) with preemptive mode <table><tr><td>Process ID</td><td>Arrival Time (AT)</td><td>Burst Time (BT)</td></tr><tr><td>P1</td><td>0</td><td>5</td></tr><tr><td>P2</td><td>1</td><td>3</td></tr><tr><td>P3</td><td>2</td><td>4</td></tr><tr><td>P4</td><td>4</td><td>1</td></tr></table>	Process ID	Arrival Time (AT)	Burst Time (BT)	P1	0	5	P2	1	4	P3	2	2	P4	4	1	Process ID	Arrival Time (AT)	Burst Time (BT)	P1	0	5	P2	1	3	P3	2	4	P4	4	1	10	L3	CO2																																							
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	b.	Explain the various operations of the processing with a neat diagram.	10	L2	CO2																																																																					
Module – 3																																																																										
Q.5	a.	Explain Dining Philosopher’s problem, illustrate using semaphore, how to handle it.	10	L3	CO3																																																																					
	b.	Considering the following example of a system, check whether the system is safe or not using Banker’s Algorithm. Also determine the sequence of if it is safe. <table><tr><td rowspan="2">Process</td><td colspan="3">Allocation</td><td colspan="3">Max</td><td colspan="3">Work Available</td></tr><tr><td>A</td><td>B</td><td>C</td><td>A</td><td>B</td><td>C</td><td>A</td><td>B</td><td>C</td></tr><tr><td>P₀</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P₁</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table>	Process	Allocation			Max			Work Available			A	B	C	A	B	C	A	B	C	P ₀	0	1	0	7	5	3	3	3	2	P ₁	2	0	0	3	2	2				P ₂	3	0	2	9	0	2				P ₃	2	1	1	2	2	2				P ₄	0	0	2	4	3	3				10	L3	CO3
Process	Allocation			Max			Work Available																																																																			
	A	B	C	A	B	C	A	B	C																																																																	
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P ₃	2	1	1	2	2	2																																																																				
P ₄	0	0	2	4	3	3																																																																				

OR					
Q.6	a.	What is deadlock? Explain the necessary conditions for the deadlock in detail.	10	L2	CO3
	b.	Illustrate with example Peterson's solution for critical section problem and prove that the mutual exclusion property is preserved.	10	L4	CO3
Module – 4					
Q.7	a.	Consider the following page reference string 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2,3,7,6,3,2,1,2,3,6. How many page faults would occur, assuming three frames for FIFO, LRU and Optimal Page Replacement?	10	L3	CO4
	b.	Write short note on : i) Internal and external fragmentation ii) Dynamic loading and linking.	10	L2	CO4
OR					
Q.8	a.	Explain contiguous memory allocation.	10	L2	CO4
	b.	Illustrate with a help of supporting diagram TLB improves the performance of demand paging.	10	L3	CO4
Module – 5					
Q.9	a.	Explain the following terms briefly. i) File attributes ii) File type	10	L2	CO5
	b.	With a neat diagram illustrate the working of various file access methods.	10	L3	CO5
OR					
Q.10	a.	Explain the following terms briefly. i) File operations ii) File system mounting	10	L2	CO5
	b.	Illustrate the various directory structures and discuss in detail.	10	L3	CO5

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22MCA13

First Semester MCA Degree Examination, Dec.2023/Jan.2024

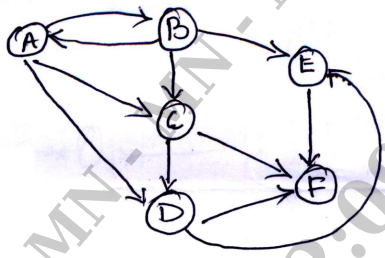
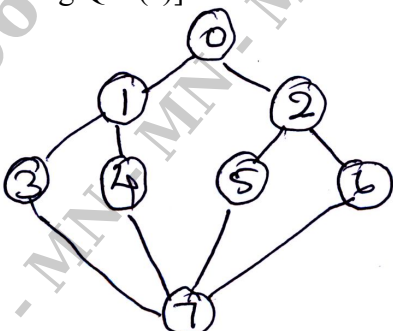
Data Structures with Algorithms

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What are data structures? Explain the classifications of data structures.	08	L1	CO1
	b.	Write a 'C' program to convert postfix to infix expression.	08	L3	CO2
	c.	Evaluate the following postfix expression using stack: 5, 6, 2, +, *, 12, 4, 1, -	04	L3	CO2
OR					
Q.2	a.	Write a 'C' program to convert infix to postfix expression using applications of stack.	10	L3	CO2
	b.	Define STACK. Write a C program to implement stack operations using arrays.	10	L3	CO2
Module – 2					
Q.3	a.	Write a 'C' program to implement tower of Honai problem using recursion and trace the output for 3 disks.	10	L3	CO2
	b.	Write a 'C' recursive functions to implement GCD of 2 numbers and generating Fibonacci sequence.	10	L3	CO2
OR					
Q.4	a.	Define circular queues. Write a 'C' program to implement circular queue operations.	10	L3	CO3
	b.	What are priority queues? Write a program to simulate priority queues with priority 1 and 2.	10	L3	CO3
Module – 3					
Q.5	a.	What are Linked lists? Write a program to implement the following options: (i) Insert a node at the beginning of the list. (ii) Delete a node at the end of the list.	10	L3	CO3
	b.	Give an account of: (i) Static and dynamic memory allocation (ii) Getnode() and freenode() operations	10	L3	CO3
OR					

Q.6	a.	Write a 'C' program to implement STACK operations using linked lists.	10	L3	CO3
	b.	Give an account of : (i) Memory management functions (ii) Array implementation of lists	10	L3	CO3
Module – 4					
Q.7	a.	Explain the array and linked representation of binary trees with suitable examples.	06	L3	CO3
	b.	Construct the binary search tree for the following array items: 40, 60, 50, 33, 55, 11	06	L3	CO3
	c.	Write a C function to create binary search tree.	08	L3	CO3
OR					
Q.8	a.	Explain binary tree traversal methods with 'C' functions and examples.	10	L3	CO3
	b.	Give an account of threaded binary trees.	10	L3	CO3
Module – 5					
Q.9	a.	Define a graph. For a graph shown in Fig.Q9(a), write the adjacency matrix and adjacency list representations.	08	L3	CO3
		 Fig.Q9(a)			
	b.	Suppose an array contains 8 elements such as 77, 33, 44, 11, 88, 22, 66, 55. Sort the array using insertion sort algorithm.	08	L3	CO4
	c.	What is hashing? Explain any two hash functions with proper examples.	06	L3	CO4
OR					
Q.10	a.	Briefly explain Breadth-First-Search (BFS) and Depth-First-Search (DFS) traversal of a graph. Also, show the BFS and DFS traversals for the following graphs. [Refer Fig.Q10(a)]	10	L3	CO4
		 Fig.Q10(a)			
	b.	Explain the working operation of Radix sort for the following set of data: 348, 143, 361, 423, 538, 128, 321, 543, 366	05	L3	CO4
	c.	Explain Address Calculation Sorting method with suitable example.	05	L3	CO4

CBCS SCHEME

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22MCA14

First Semester MCA Degree Examination, Dec.2023/Jan.2024

Computer Networks

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define Computer Networks. Explain the fundamental characteristics of data communication.	6	L1	CO1
	b.	With a neat diagram, discuss the categories of topology.	8	L1	CO1
	c.	With the help of a neat diagram, explain the different types of communication between two devices.	6	L2	CO1
OR					
Q.2	a.	Explain the layers of ISO OSI model with a neat diagram.	10	L2	CO1
	b.	How networks are categorized? Explain LAN, WAN and MAN.	10	L2	CO1
Module – 2					
Q.3	a.	A device sends 265 kbps over a noiseless channel with a bandwidth of 20KHz. How many signal levels do we need?	4	L3	CO2
	b.	List and explain the causes of transmission impairments?	12	L2	CO2
	c.	Explain the truss bandwidth and throughput.	4	L2	CO2
OR					
Q.4	a.	Explain sampling and Quantization of pulse code modulation?	8	L2	CO2
	b.	Discuss frequency shift keying in brief.	4	L2	CO2
	c.	For the following bit pattern 01001110, apply NRZ – L, NRZ – I, Manchester and differential Manchester encoder schemes.	8	L3	CO2
Module – 3					
Q.5	a.	What is Multiplexing? With the help of a neat diagram? Explain frequency Division Multiplexing (FDM).	10	L2	CO3
	b.	Explain frequency Hopping Spread Spectrum (FHSS).	10	L2	CO3
OR					
Q.6	a.	List all the characteristics of Virtual – circuit network.	5	L2	CO3
	b.	Explain in brief the three phases of a circuit switched network with a neat diagram.	9	L2	CO3
	c.	What is the role of routing table in datagram network? Explain.	6	L2	CO3

Module – 4					
Q.7	a.	Solve CRC encoder and decoder considering the values for dataword = 1001 and divisor = 1011.	8	L3	CO4
	b.	Explain simple parity check code.	6	L2	CO4
	c.	Discuss the following : i) Error Detection ii) Error correction.	6	L2	CO4
OR					
Q.8	a.	Suppose our data is a list of five 4-bit numbers that we want to send the set of numbers (7, 11, 12, 0, 6) to a destination. Apply checksum process at the sender side and at the receiver side.	10	L3	CO4
	b.	How Cyclic codes can understand, using polynomials? Explain with proper example.	10	L2	CO4
Module – 5					
Q.9	a.	Explain how stop and wait protocol helps in achieving flow control.	10	L2	CO4
	b.	Discuss the working of Go-Back-N Automatic Report Request protocol.	10	L2	CO4
OR					
Q.10	a.	Explain the working of selective Report ARQ.	10	L2	CO4
	b.	Discuss the working of Point-To-Point protocol.	10	L2	CO4

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First Semester MCA Degree Examination, Dec.2023/Jan.2024

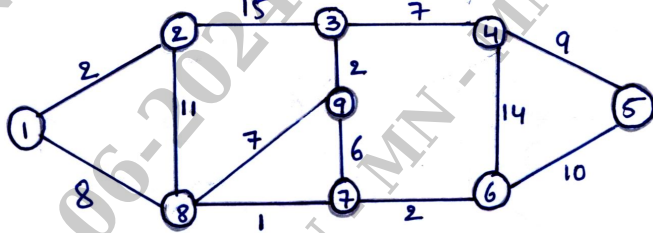
Design and Analysis of Algorithm

Time: 3 hrs.

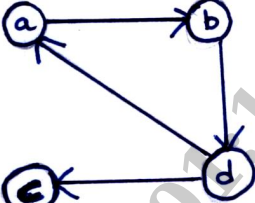
Max. Marks: 100

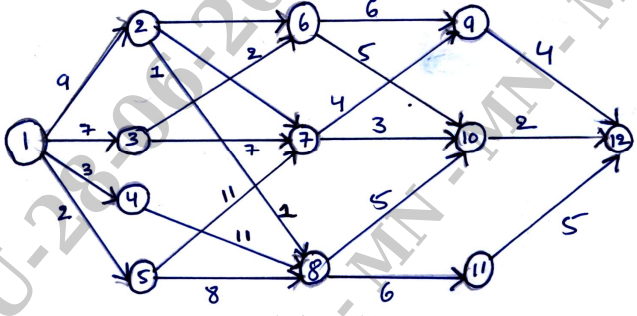
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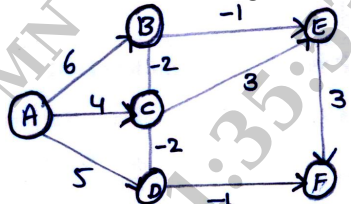
Module – 1			M	L	C											
Q.1	a.	Explain the mathematical analysis of recursive algorithm with an example of finding the factorial of 'n' number.	10	L2	CO1											
	b.	Explain asymptotic notation with example.	10	L2	CO1											
OR																
Q.2	a.	What is an algorithm? Explain algorithm specification and analysis frame work with example.	10	L2	CO1											
	b.	Explain the mathematical analysis of non recursive algorithm with an example.	10	L2	CO1											
Module – 2																
Q.3	a.	Write an algorithm for merge sort, find the time complexity of merge sort.	10	L2	CO1											
	b.	Explain divide and conquer, explain maximum and minimum using divide and conquer technique.	10	L2	CO1											
OR																
Q.4	a.	Write an algorithm for quick sort and analyze it efficiency.	10	L2	CO1											
	b.	Write an algorithm for heap sort with example.	10	L2	CO2											
Module – 3																
Q.5	a.	Write the Prim's algorithm, apply this algorithm to following graph in Fig.Q.5(a) to construct minimum spanning tree.  Fig.Q.5(a)	10	L2	CO1											
	b.	Explain Dijkstra's algorithm with example.	10	L2	CO1											
OR																
Q.6	a.	Explain knapsack problem by greedy method with example.	10	L2	CO1											
	b.	Find the Huffman code for the following data by obtaining Huffman tree. <table border="1" data-bbox="507 1877 1070 1951"> <tr> <td>Character</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr> <tr> <td>Probability</td><td>0.11</td><td>0.40</td><td>0.16</td><td>0.09</td><td>0.24</td></tr> </table>	Character	A	B	C	D	E	Probability	0.11	0.40	0.16	0.09	0.24	10	L2
Character	A	B	C	D	E											
Probability	0.11	0.40	0.16	0.09	0.24											

Module – 4

Q.7	a. Write an algorithm to complete transitive closure for the given graph, and obtain the transitive closure for the given graph show in Fig.Q.7(a) using Warshall's algorithm.  Fig.Q.7(a)	10	L2	CO2
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	b. Define multistage graph problem, determine the minimum cost path from source (s) to sink (T) for the graph in Fig.Q.7(b) using forward approach.  Fig.Q.7(b)	10	L2	CO1
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OR

Q.8	a. Solve the below instance of bellmen-ford algorithm.  Fig.Q.8(a)	10	L2	CO1
	b. Explain travelling sales person problem with example.	10	L2	CO1

Module – 5

Q.9	a. Explain N-Queue problem using back-tracking method.	10	L2	CO3
	b. Define the following: i) Class P ii) Class NP.	10	L2	CO3

OR

Q.10	a. Apply back tracking technique to solve the below instance of the subset sum problem. $S = \{1, 3, 4, 6\}$ $d = 7$.	10	L2	CO3
	b. Define the following: i) NP complete problem ii) NP hard problem.	10	L2	CO3

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First Semester MCA Degree Examination, Dec.2023/Jan.2024

Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define the meaning of research. Discuss the objectives and motivation in research.	10	L2	CO1
	b.	Differentiate between researcher methods versus research methodology.	10	L2	CO1
OR					
Q.2	a.	Briefly explain the criteria of good research.	10	L1	CO1
	b.	Explain the problems encountered by Researcher in India.	10	L1	CO1
Module – 2					
Q.3	a.	What is a research problem? Explain the issues to be resolved by researcher in formulating a research problem.	10	L1	CO2
	b.	What are the aspects the researcher should follow and examine the literature reviewed.	10	L1	CO2
OR					
Q.4	a.	What is literature review? Explain the process to review literature.	10	L2	CO2
	b.	Explain the procedure to develop a conceptual framework in research.	10	L3	CO2
Module – 3					
Q.5	a.	What is the meaning of research design? Explain the important concepts relating to research design.	10	L2	CO3
	b.	List and explain few important experimental design in research.	10	L2	CO3
OR					
Q.6	a.	What are the characteristics of a good sample design?	10	L1	CO3
	b.	Explain the different types of sampling design.	10	L2	CO3
	c.	The following are the number of departmental stores in 15 cities = 35, 17, 10, 32, 70, 28, 26, 19, 26, 66, 37, 44, 33, 29 and 28. If we want to select a sample of 10 stores using cities as clusters and selecting within clusters proportional to size, how many stores from each city should be chosen. (Use a starting point of 10).			
Module – 4					
Q.7	a.	Enumerate the different methods of data collection in research.	10	L2	CO4
	b.	What are the techniques of interpretation?	10	L2	CO4
OR					
Q.8	a.	What is the significance of report writing? Explain the steps in writing report.	10	L2	CO4
	b.	What are the precautions that should be taken while writing research problem.	10	L2	CO4
Module – 5					
Q.9	a.	What is IP? Explain the different types of IPs.	10	L2	CO5
	b.	Why do you think IPR is important in recent times?	10	L3	CO5
OR					
Q.10		Write short notes on (any two) :			
	a.	Indian Patent Act 1970	10	L1	CO5
	b.	Copyright Act 1957.	10	L1	CO5